

Lecture 08. Model-free RL

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29.04.2026

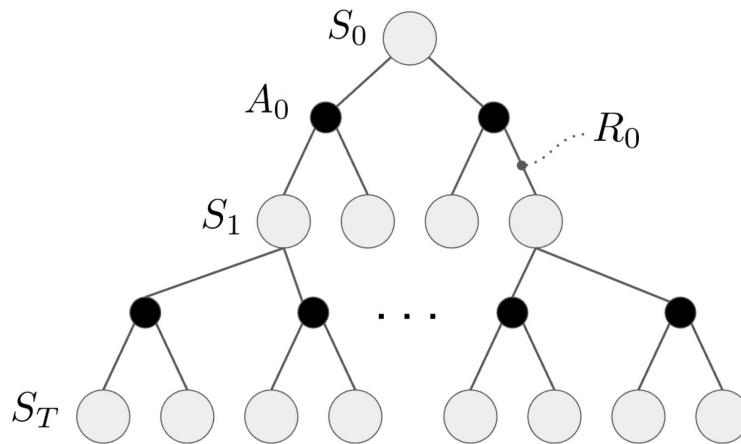
Outline

- Model-based and model-free RL
- Recap: solving mdp with dynamic programming
- Model-free prediction
 - Monte-Carlo vs. TD
- Model-free control
 - SARSA
 - Q-Learning
- Exploration / exploitation tradeoff

Model-free vs. Model-based RL

Model-based RL

1. Know **the complete dynamics** of MDP
2. Can plan ahead
3. Do not need actual experiences to estimate return

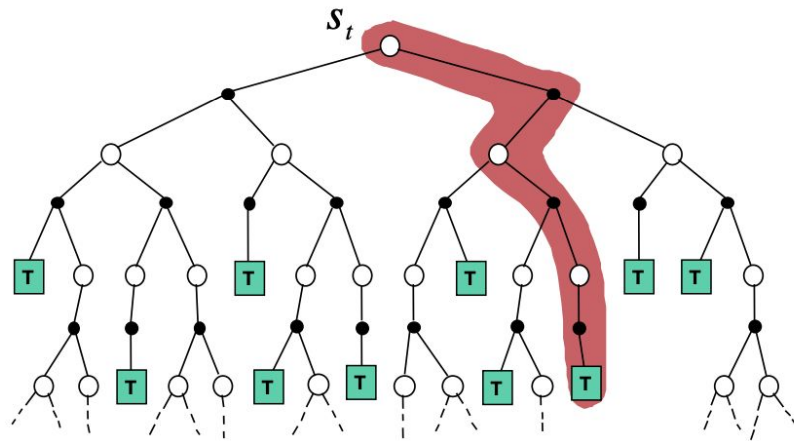


$$v_{\pi}(s) = \sum_a \pi(a | s) \sum_{r, s'} p(r, s' | s, a) [r + \gamma v_{\pi}(s')]$$

Model-free vs. Model-based RL

Model-free RL

1. MDP inner structure is unknown
2. Can only try stuff and estimate
3. Need samples of past experiences to learn



Recap: model-based learning

State and action-value functions

$$G_t = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-1} R_T$$

$$v_\pi(s) \triangleq \mathbb{E}_\pi [G_t \mid S_t = s]$$

$$= \mathbb{E}_\pi [R_t + \gamma G_{t+1} \mid S_t = s]$$

$$= \sum_a \pi(a \mid s) \sum_{r, s'} p(r, s' \mid s, a) \left[r + \gamma \mathbb{E}_\pi [G_{t+1} \mid S_{t+1} = s'] \right]$$

$$= \sum_a \pi(a \mid s) \sum_{r, s'} p(r, s' \mid s, a) [r + \gamma v_\pi(s')]$$

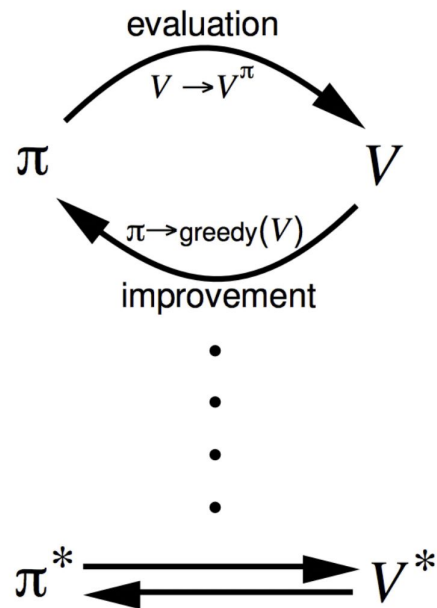
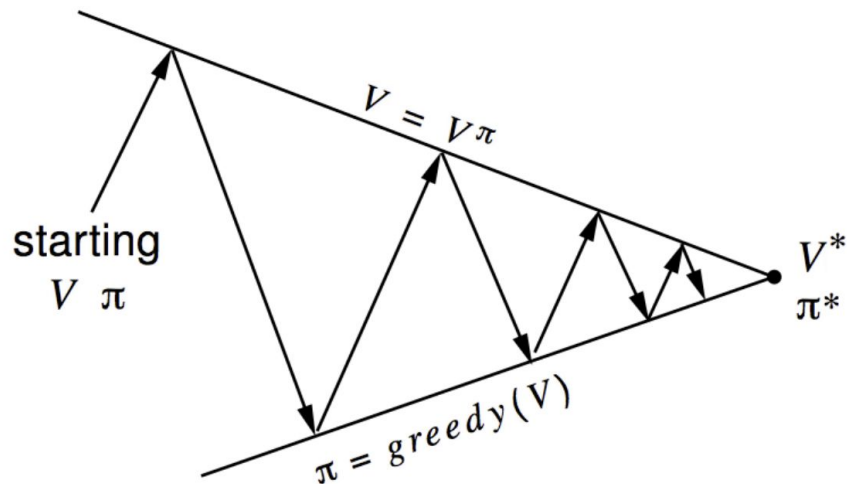
State and action-value functions

$$G_t = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-1} R_T$$

$$\begin{aligned} q_\pi(s, a) &= \mathbb{E}_\pi [G_t \mid S_t = s, A_t = a] \\ &= \mathbb{E}_\pi [R_t + \gamma G_{t+1} \mid S_t = s, A_t = a] \\ &= \sum_{r, s'} p(r, s' \mid s, a) \left[r + \gamma \mathbb{E}_\pi [G_{t+1} \mid S_{t+1} = s'] \right] \\ &= \sum_{r, s'} p(r, s' \mid s, a) [r + \gamma v_\pi(s')] \end{aligned}$$

Policy iteration

1. Policy evaluation - given policy π , estimate V_π
2. Policy improvement - improve π greedily



Policy iteration: Bellman equations

Bellman expectation equations

$$\begin{aligned}v_{\pi}(s) &= \sum_a \pi(a | s) \sum_{r, s'} p(r, s' | s, a) [r + \gamma v_{\pi}(s')] \\&= \mathbb{E}_{\pi} [R_t + \gamma v_{\pi}(S_{t+1}) | S_t = s]\end{aligned}$$

$$\begin{aligned}q_{\pi}(s, a) &= \sum_{r, s'} p(r, s' | s, a) [r + \gamma v_{\pi}(s')] \\&= \sum_{r, s'} p(r, s' | s, a) \left[r + \gamma \sum_{a'} \pi(a' | s') q_{\pi}(s', a') \right]\end{aligned}$$

How to estimate V and Q
functions for **a given**
policy π

Policy iteration: Bellman equations

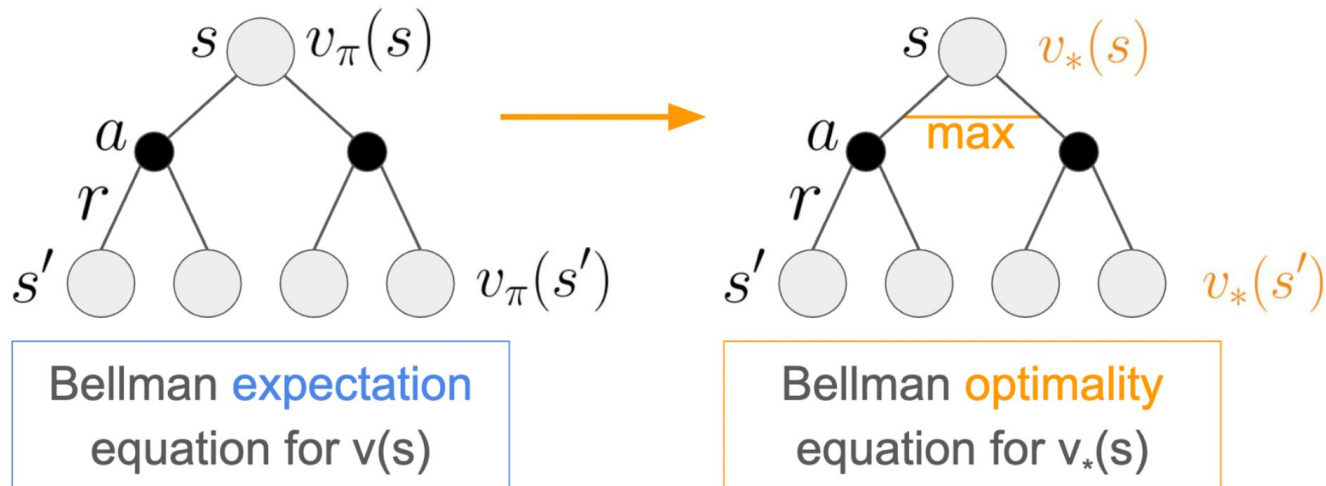
Bellman optimality equations

Optimal strategy

$$\pi \geq \pi' \iff v_{\pi}(s) \geq v_{\pi'}(s) \quad \forall s$$

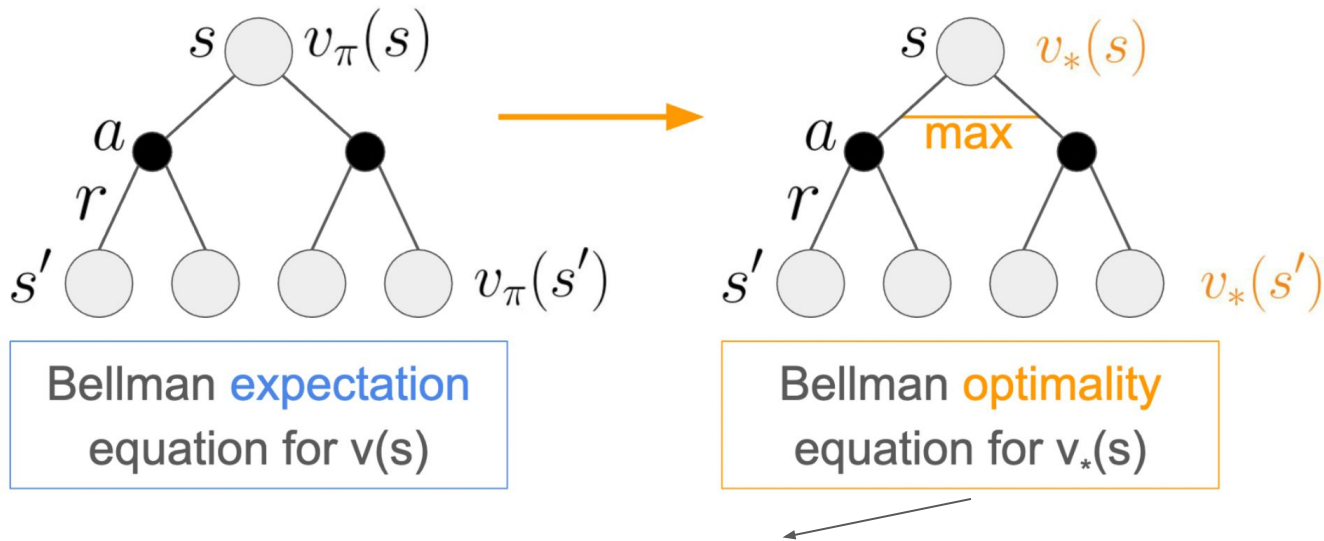
Best policy π_* is better or equal to any other policy

Policy iteration: Bellman equations



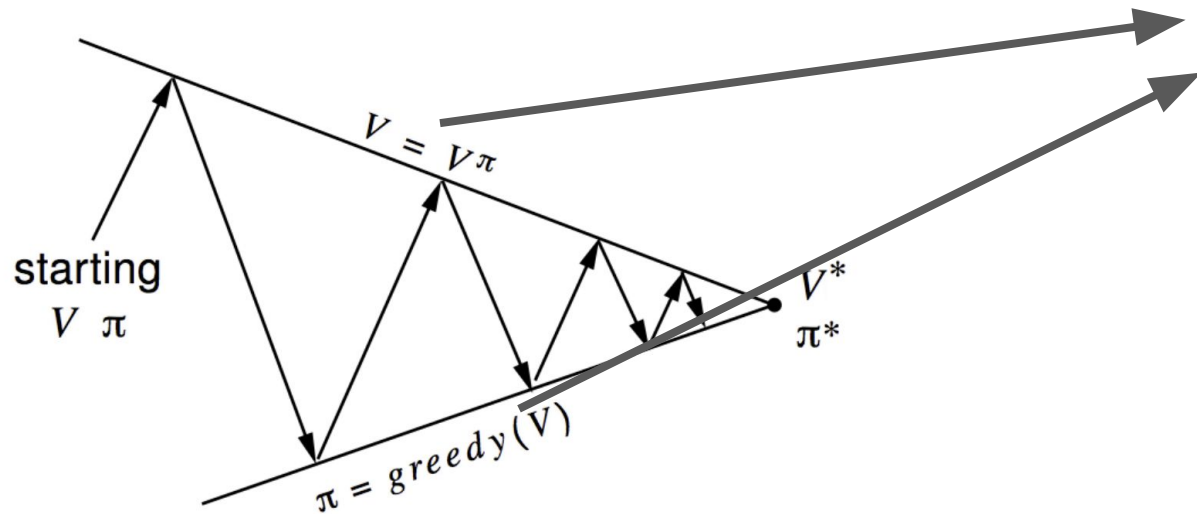
$$\begin{aligned} v_\pi(s) &= \sum_a \pi(a | s) \sum_{r, s'} p(r, s' | s, a) [r + \gamma v_\pi(s')] \\ &= \mathbb{E}_\pi [R_t + \gamma v_\pi(S_{t+1}) | S_t = s] \end{aligned}$$

Policy iteration: Bellman equations



$$\begin{aligned} v_*(s) &= \max_a \sum_{r, s'} p(r, s' | s, a) [r + \gamma v_*(s')] \\ &= \max_a \mathbb{E} [R_t + \gamma v_*(S_{t+1}) | S_t = s, A_t = a] \end{aligned}$$

Policy iteration: convergence



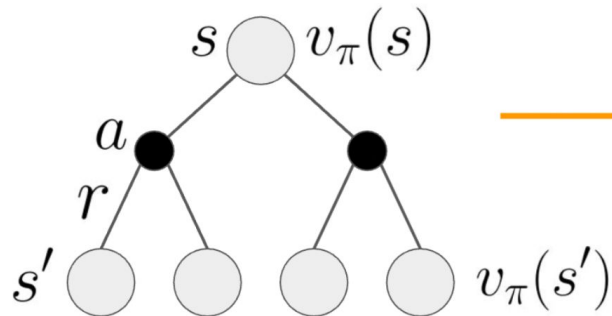
**Contraction
operators (both)**

**Hence, a fixed point
exists**

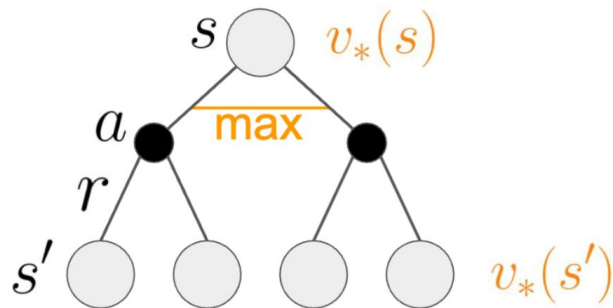
Model-free prediction

Model-free prediction

Q.: How to estimate V , Q functions for a given policy, **without MDP dynamics**?



Bellman **expectation**
equation for $v(s)$

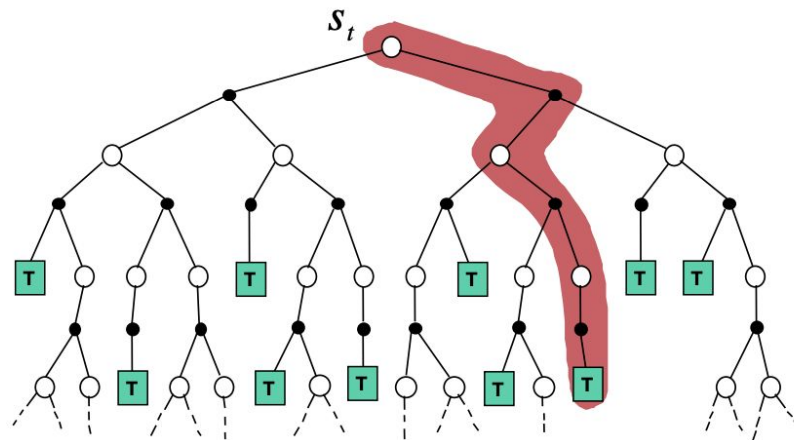


Bellman **optimality**
equation for $v_*(s)$

Any problems?

Model-free: Learning from trajectories

- Sample a lot of sessions from our current π
- Look at the cumulative returns for each state
- Average every visit



Monte-Carlo Policy Evaluation

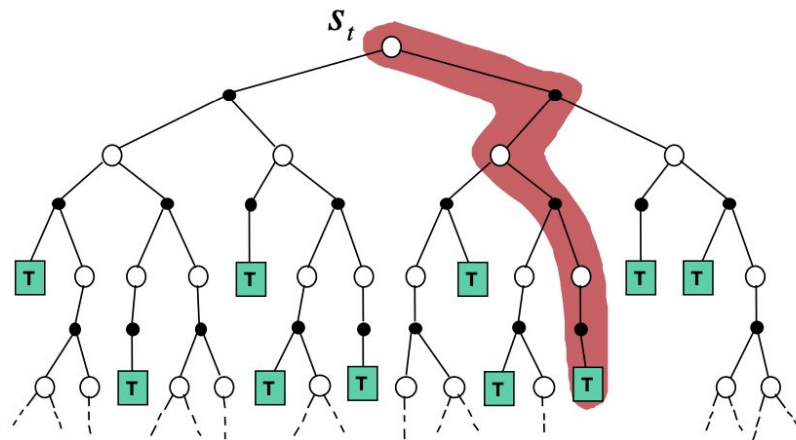
- Sample a lot of sessions from our current π
- Every time state s is visited

$$N(s) \leftarrow N(s) + 1$$

$$S(s) \leftarrow S(s) + G_t$$

$$V(s) = S(s)/N(s)$$

$$V(s) \rightarrow v_{\pi}(s) \text{ as } N(s) \rightarrow \infty$$



Incremental MC Policy Evaluation

Running mean updates

$$\begin{aligned}\mu_k &= \frac{1}{k} \sum_{j=1}^k x_j \\ &= \frac{1}{k} \left(x_k + \sum_{j=1}^{k-1} x_j \right) \\ &= \frac{1}{k} (x_k + (k-1)\mu_{k-1}) \\ &= \mu_{k-1} + \frac{1}{k} (x_k - \mu_{k-1})\end{aligned}$$

Incremental MC Policy Evaluation

Running mean updates

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$$N(S_t) \leftarrow N(S_t) + 1$$

$$V(S_t) \leftarrow V(S_t) + \frac{1}{N(S_t)} (G_t - V(S_t))$$

Incremental MC Policy Evaluation

Running mean updates

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Constant LR is useful to forget old episodes (for non-stationary setup):

$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

Incremental MC Policy Evaluation

Running mean updates

Q.: Any problems?

$$\begin{aligned}\mu_k &= \frac{1}{k} \sum_{j=1}^k x_j \\ &= \frac{1}{k} \left(x_k + \sum_{j=1}^{k-1} x_j \right) \\ &= \frac{1}{k} (x_k + (k-1)\mu_{k-1}) \\ &= \mu_{k-1} + \frac{1}{k} (x_k - \mu_{k-1})\end{aligned}$$

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Constant LR is useful to forget old episodes (for non-stationary setup):

$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

Temporal Difference (TD) updates

Idea:

1. Monte-Carlo updates a $V(s)$ guess towards **a sample from true $V(s)$ distribution**

$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

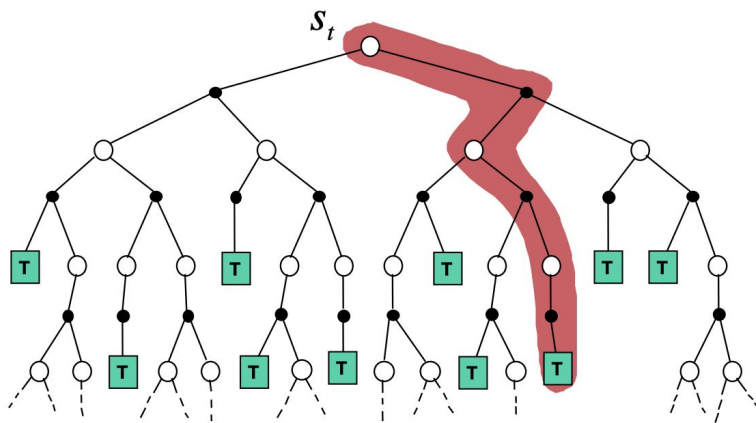
2. Let's rollout only one step ahead and update a guess towards **a slightly more precise guess**

$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

Temporal Difference (TD) updates

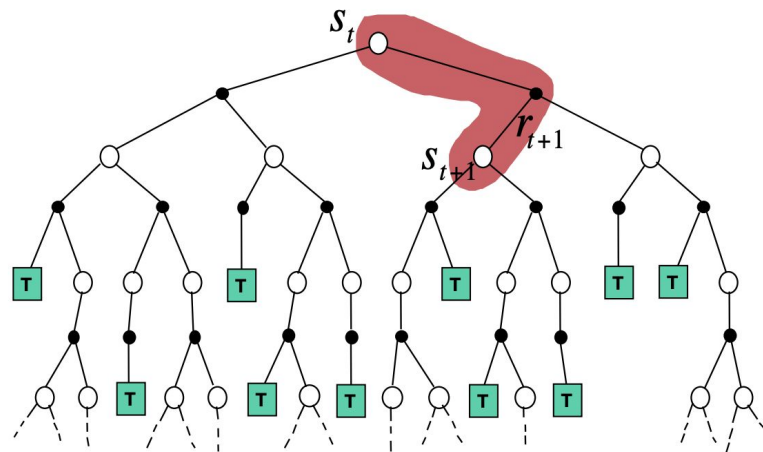
Monte-Carlo backup

$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$



Temporal Difference backup

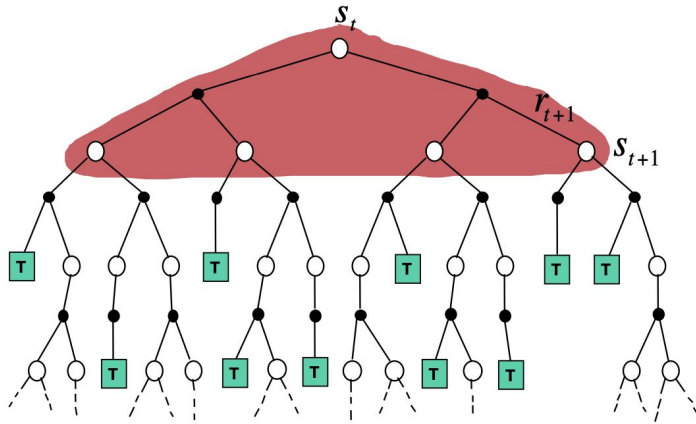
$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$



Temporal Difference (TD) updates

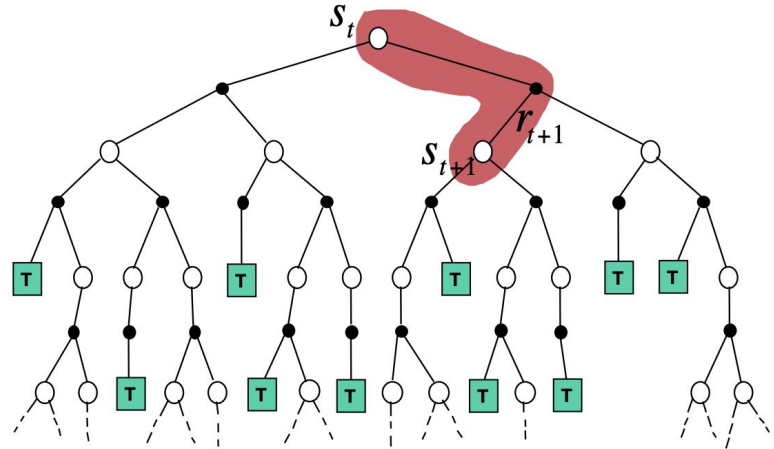
Dynamic Programming backup

$$V(S_t) \leftarrow \mathbb{E}_{\pi} [R_{t+1} + \gamma V(S_{t+1})]$$



Temporal Difference backup

$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$



TD vs. MC

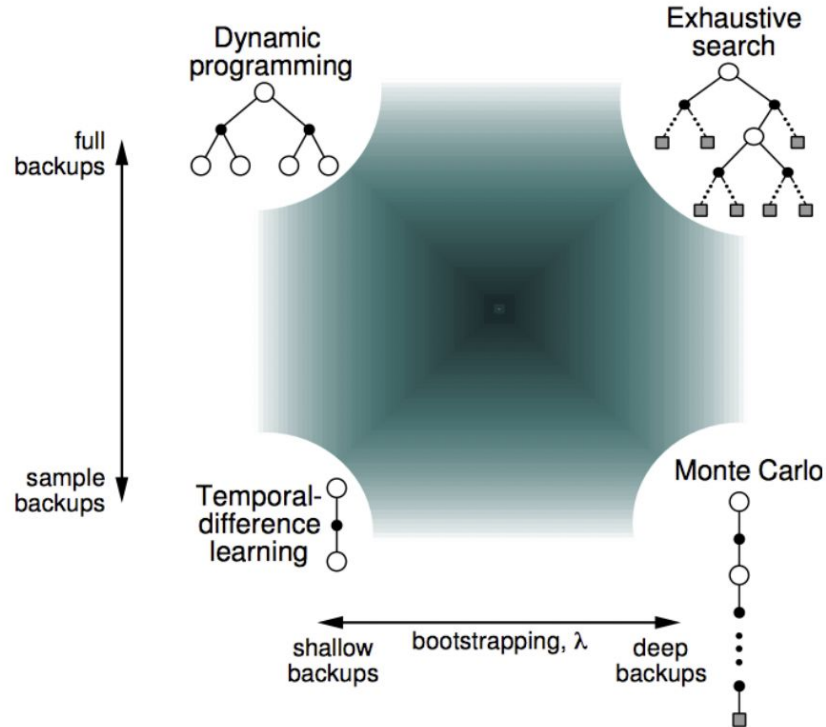
Monte-Carlo:

- honest sampling
- unbiasedly estimates expectation
- converges to MSE solution with G_t targets

TD:

- bootstrapping
- biased estimator (as “target” involves error)
- converges to solution of max likelihood Markov model

Value-based methods. Unified view.



TD(n)

- Use n-step rollouts instead of 1-step
- Slightly more accurate bootstraps

Q.: what is $TD(inf)$?

TD(n)

- Use n-step rollouts instead of 1-step
- Slightly more accurate bootstraps

$$n = 1 \quad (TD) \quad G_t^{(1)} = R_{t+1} + \gamma V(S_{t+1})$$

$$n = 2 \quad G_t^{(2)} = R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2})$$

$$\vdots \quad \vdots$$

$$n = \infty \quad (MC) \quad G_t^{(\infty)} = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-1} R_T$$

$$V(S_t) \leftarrow V(S_t) + \alpha \left(G_t^{(n)} - V(S_t) \right)$$

Model-free control

Model-free control

Model-free prediction

- ***Estimate*** the value function of unknown MDP

Model-free control

- ***Optimize*** the value function of unknown MDP

Model-free control

Q.: What to learn?

$V(S)$ or $Q(S, A)$

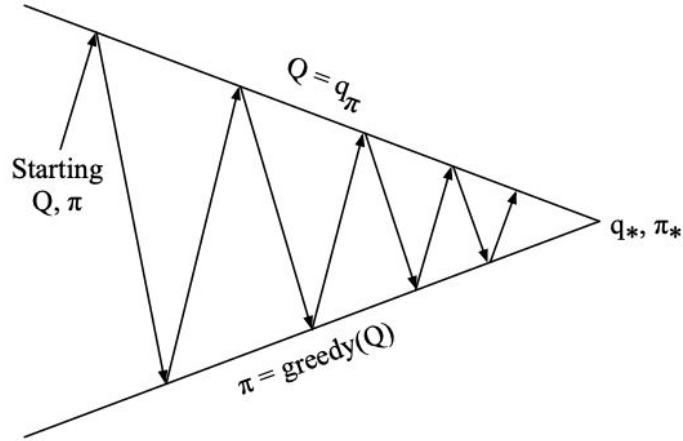
Model-free control

Q.: What to learn?

$V(S)$ or $Q(S, A)$

$V(S)$ is useless for action prediction w/o model dynamics

Policy Iteration with Q-function



1. Policy evaluation
estimate Q using MC sampling
2. Policy improvement
greedy improvement

Exploration / exploitation tradeoff

- Two doors
- Try left door, $V(\text{left}) = 0$
- Try right door, $V(\text{right}) = 1$

Q.: Is the right door optimal?



"Behind one door is tenure - behind the other is flipping burgers at McDonald's."

Exploration / exploitation tradeoff

- Two doors
- Try left door, $V(\text{left}) = 0$
- Try right door, $V(\text{right}) = 1$

Q.: Is the right door optimal?

- *Stochasticity in environment*
- *Need to ensure continual exploration of options*



"Behind one door is tenure - behind the other is flipping burgers at McDonald's."

Exploration / exploitation tradeoff

Epsilon-greedy exploration

- with probability epsilon, pick random action uniformly
- with probability (1 - epsilon), pick current best choice

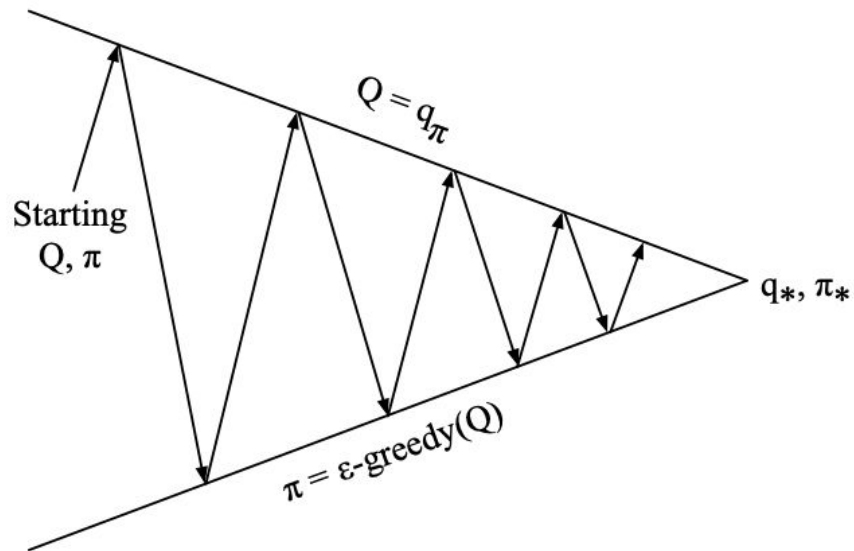
$$\pi(a|s) = \begin{cases} \epsilon/m + 1 - \epsilon & \text{if } a^* = \operatorname{argmax}_{a \in \mathcal{A}} Q(s, a) \\ \epsilon/m & \text{otherwise} \end{cases}$$

Epsilon-greedy policy improvement

$$\begin{aligned}q_{\pi}(s, \pi'(s)) &= \sum_{a \in \mathcal{A}} \pi'(a|s) q_{\pi}(s, a) \\&= \epsilon/m \sum_{a \in \mathcal{A}} q_{\pi}(s, a) + (1 - \epsilon) \max_{a \in \mathcal{A}} q_{\pi}(s, a) \\&\geq \epsilon/m \sum_{a \in \mathcal{A}} q_{\pi}(s, a) + (1 - \epsilon) \sum_{a \in \mathcal{A}} \frac{\pi(a|s) - \epsilon/m}{1 - \epsilon} q_{\pi}(s, a) \\&= \sum_{a \in \mathcal{A}} \pi(a|s) q_{\pi}(s, a) = v_{\pi}(s)\end{aligned}$$

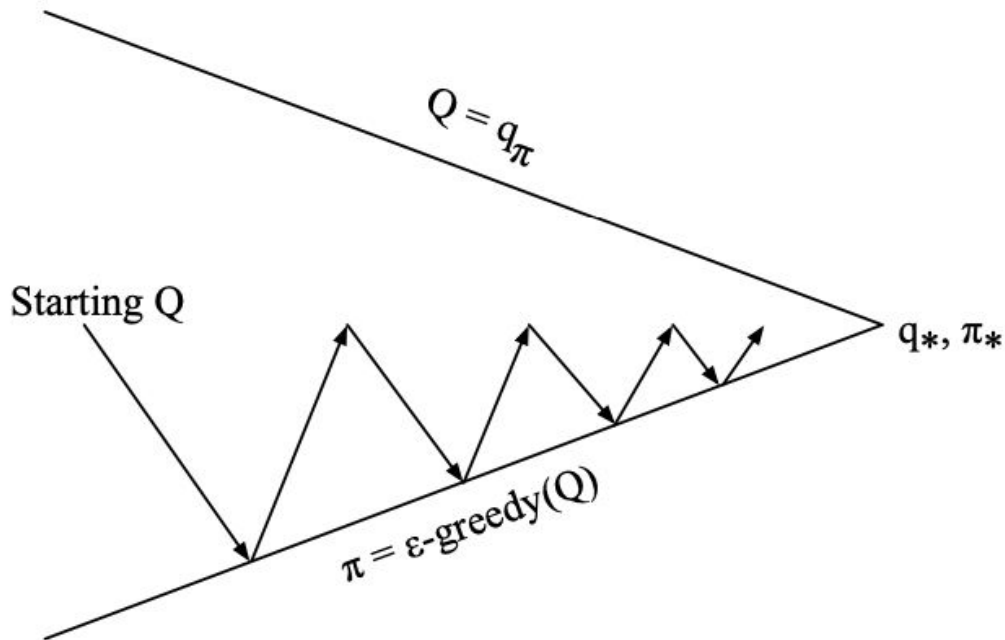
$$v_{\pi'}(s) \geq v_{\pi}(s)$$

Monte-Carlo Policy Iteration



1. Policy evaluation
estimate Q using MC sampling
2. Policy improvement
 ϵ -greedy improvement

Monte-Carlo Control



Just like in value iteration

on every episode (!)

1. *Approx. policy evaluation*
2. *Policy improvement*

Model Free Control: SARSA

Basically, SARSA is

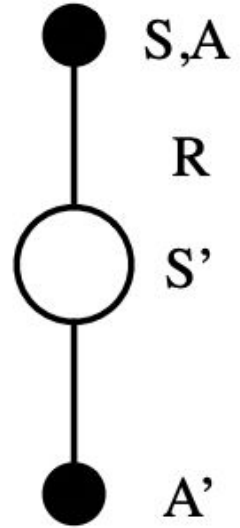
- One-step Temporal Difference Policy Evaluation
- Epsilon-greedy Policy Improvement

Model Free Control: SARSA

Basically, SARSA is

- One-step Temporal Difference Policy Evaluation
- Epsilon-greedy Policy Improvement

$$Q(S, A) \leftarrow Q(S, A) + \alpha (R + \gamma Q(S', A') - Q(S, A))$$



Model Free Control: SARSA

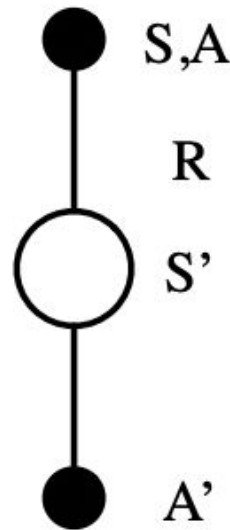
Basically, SARSA is

- One-step Temporal Difference Policy Evaluation
- Epsilon-greedy Policy Improvement

$$Q(S, A) \leftarrow Q(S, A) + \alpha (R + \gamma Q(S', A') - Q(S, A))$$



Policy evaluation



Sample from experience

Model Free Control: n-step SARSA

$$n = 1 \quad (\text{Sarsa}) \quad q_t^{(1)} = R_{t+1} + \gamma Q(S_{t+1})$$

$$n = 2 \quad q_t^{(2)} = R_{t+1} + \gamma R_{t+2} + \gamma^2 Q(S_{t+2})$$

$$\vdots$$

$$n = \infty \quad (\text{MC}) \quad q_t^{(\infty)} = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-1} R_T$$

Use n-step partial rollouts and TD(n) estimates

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left(q_t^{(n)} - Q(S_t, A_t) \right)$$

Off-policy and on-policy learning

Off-policy vs. on-policy

On-policy learning

- Training episodes are sampled from the current policy we are optimizing

Off-policy learning

- Training episodes are taken elsewhere (e.g. old policy, external policy, some data with unknown origin)

Off-policy model-free control: Q-Learning

Key idea: optimize greedy policy

- Our policy is greedy (or eps-greedy) w.r.t. current q-values
- Hence, no A' is required in (S, A, R, S', A') updates

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha (R_{t+1} + \gamma Q(S_{t+1}, A') - Q(S_t, A_t))$$

Off-policy model-free control: Q-Learning

Key idea: optimize greedy policy

- Our policy is greedy (or eps-greedy) w.r.t. current q-values
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$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha (R_{t+1} + \gamma Q(S_{t+1}, A') - Q(S_t, A_t))$$

$$\pi(S_{t+1}) = \operatorname{argmax}_{a'} Q(S_{t+1}, a')$$

$$Q(S, A) \leftarrow Q(S, A) + \alpha \left(R + \gamma \max_{a'} Q(S', a') - Q(S, A) \right)$$

Off-policy model-free control: Q-Learning

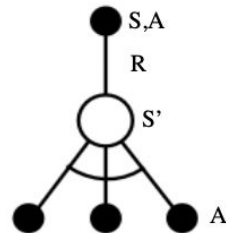
Key idea: optimize greedy policy

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- Hence, no A' is required in (S, A, R, S', A') updates

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha (R_{t+1} + \gamma Q(S_{t+1}, A') - Q(S_t, A_t))$$

$$\pi(S_{t+1}) = \operatorname{argmax}_{a'} Q(S_{t+1}, a')$$

$$Q(S, A) \leftarrow Q(S, A) + \alpha \left(R + \gamma \max_{a'} Q(S', a') - Q(S, A) \right)$$



Off-policy model-free control: EV-SARSA

EV-SARSA (Expected Value SARSA)

Use current policy to get expectation of $V(s')$

Q-learning: greedy policy // EV-SARSA: some policy

$$\hat{Q}(s, a) = r(s, a) + \gamma \underbrace{E_{a_i \sim \pi(a|s')} Q(s', a_i)}$$


“Target” Q from trajectory

Off-policy vs. on-policy

On-policy

- Agent trains on experience generated with its own policy
- Can't learn off-policy

Examples:

- Cross-entropy method
- SARSA

Off-policy

- Agent trains on any kind of experience
- Can still learn on-policy

Examples:

- Q-learning
- EV-SARSA

Outro

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 - SARSA
 - Q-Learning
- Exploration / exploitation tradeoff

Acknowledgements

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Questions?